640UD v3.0.E3
High Speed Debug Interface

Apr 29, 2026

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硬件连接

以下两个转接板都可以使用，ATUSB-HSBB-PCB支持最高6MHz的SPI通信。

Part Number: THW1027

PROMATE D21 MAXTOUCH USB BRIDGE PCB ☆

User Guide: PDF Promate_D21_User_Guide_20200611.pdf

In Stock Now : **7**

When can I get more? ⓘ

CNY/unit: **¥683.48**

View Purchasing Options

Overview Documentation

Overview

Description Features Package Contents System Requirements

This maXTouch USB Bridge product provides an interface to connect the I2C or SPI compatible based maXTouch devices to PC-based applications through a USB port.

The D21 bridge board significantly improves data throughput, increasing test and production efficiency. This board is twice as fast as other maXTouch bridge boards (ATUSB-PCB-80146 and ATUSB-I2C-AUTO-PCB).

Part Number: ATUSB-HSBB-PCB

MAXTOUCH HIGH SPEED BRIDGE BOARD ☆

In Stock Now : **1**

When can I get more? ⓘ

CNY/unit: **¥517.72**

View Purchasing Options

Overview Documentation

Overview

Description Features Package Contents System Requirements

This maXTouch USB Bridge product is an ARM-based interface board to convert I2C or SPI and debug signals to USB and communicate with the PC-based applications. This product is designed to be used with Microchip Automotive maXTouch touchscreen controllers.

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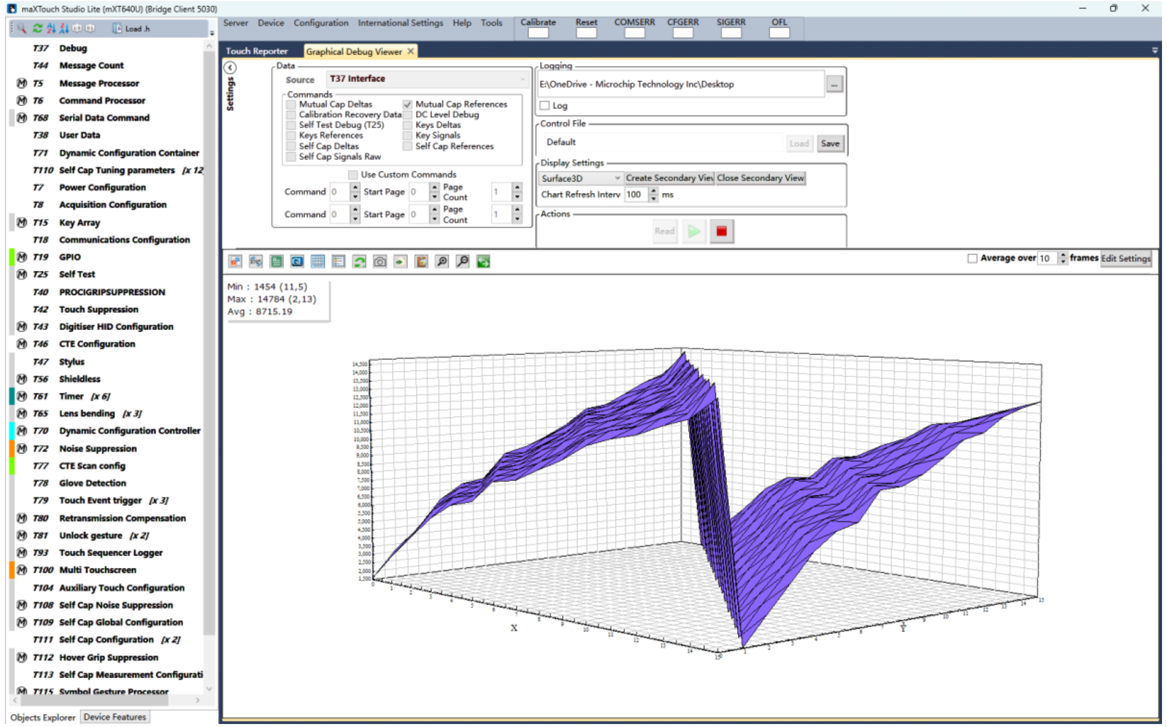
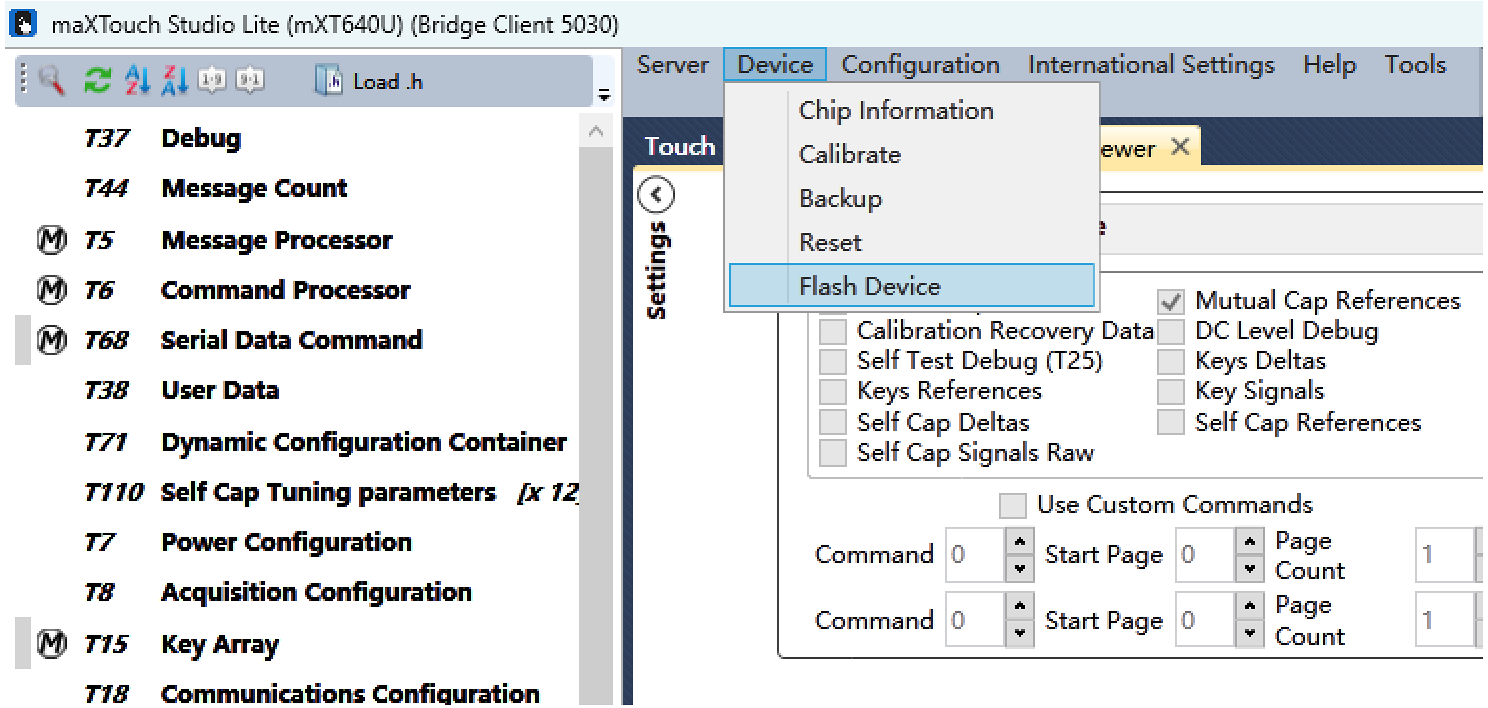
软件安装

- 下载链接
  - https://ww1.microchip.com/downloads/aemDocuments/documents/HMID/ProductDocuments/SoftwareTools/maXTouchStudio-Lite-3.1.3200.zip
- 软件使用参考说明文档
  - 附件



FW更新

- **640UD支持高速SPI接口的FW版本为V3.0.E3**
  - Enc文件：ATMXT640UD_0x17_3.0.E3_PROD.enc
  - 参考附件
- **更新步骤一**
  - 硬件和软件连接好之后，点击Device下面的Flash Device。



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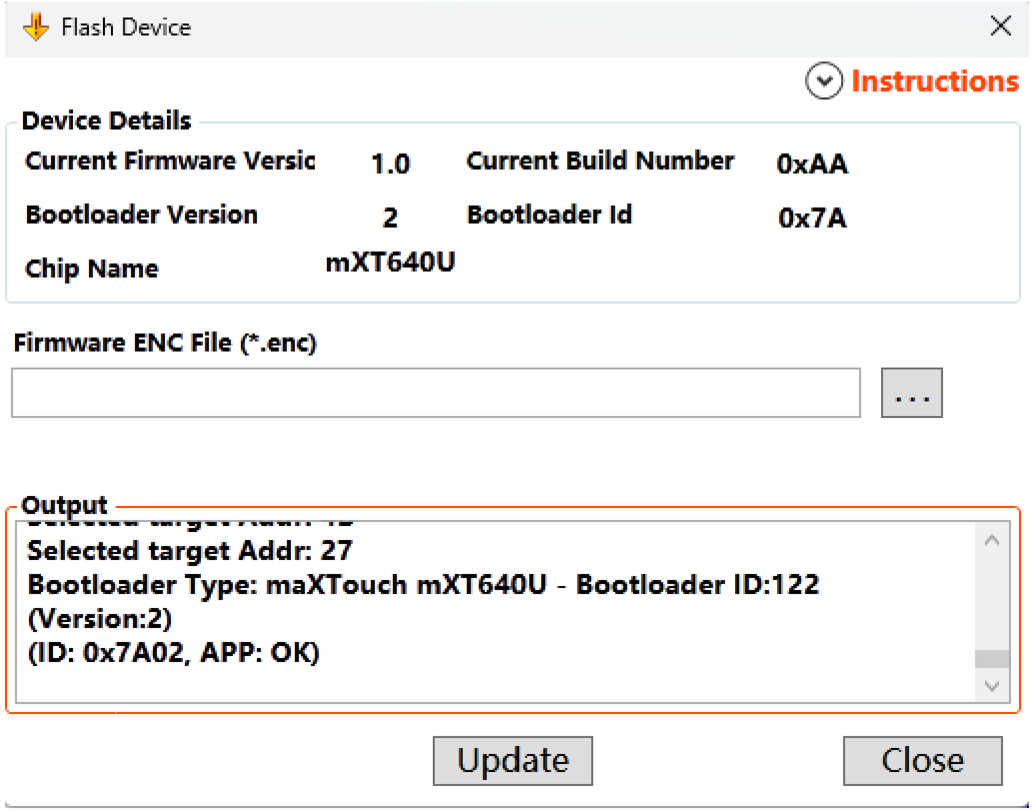
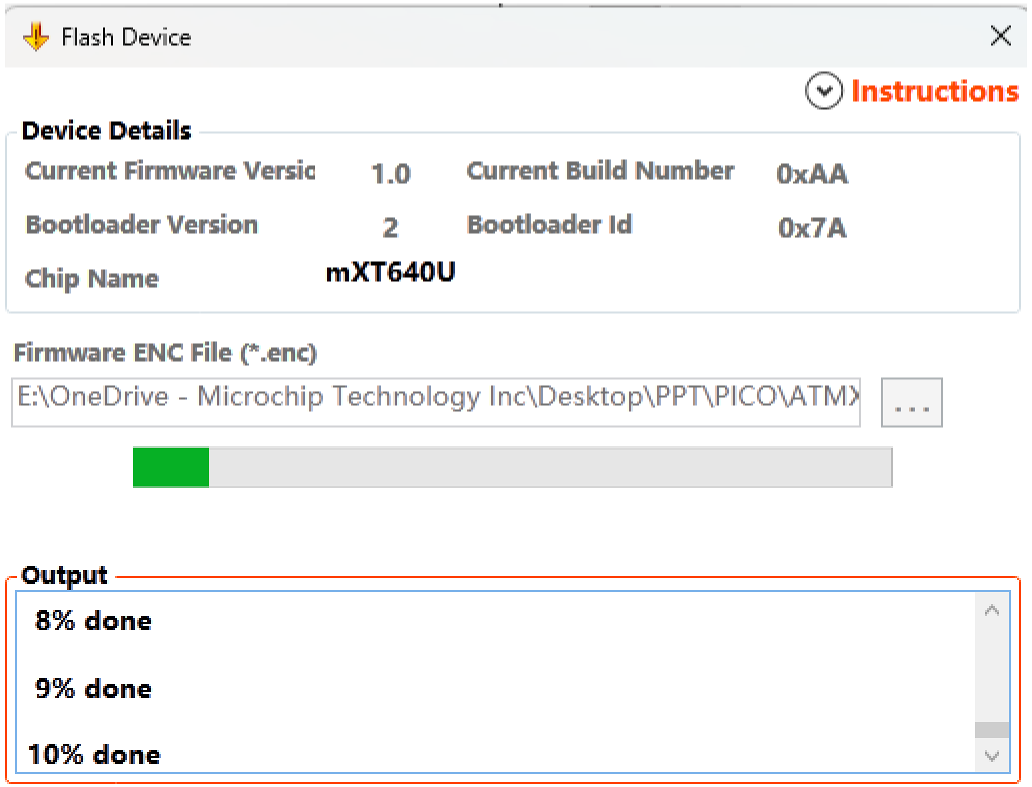
FW更新

- **步骤二**
  - 在弹出的界面选择enc文件
  - 然后点击Update



- **步骤三**
  - 等待Update完成



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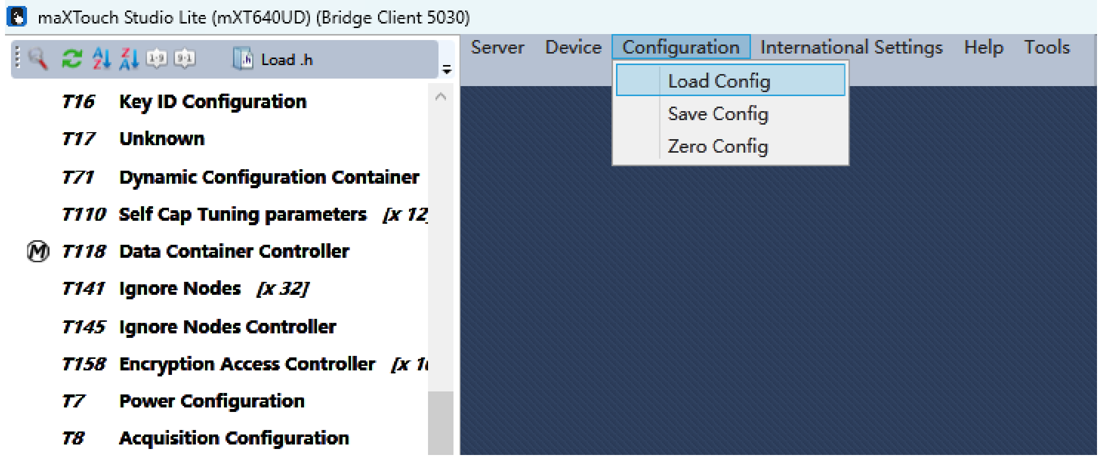
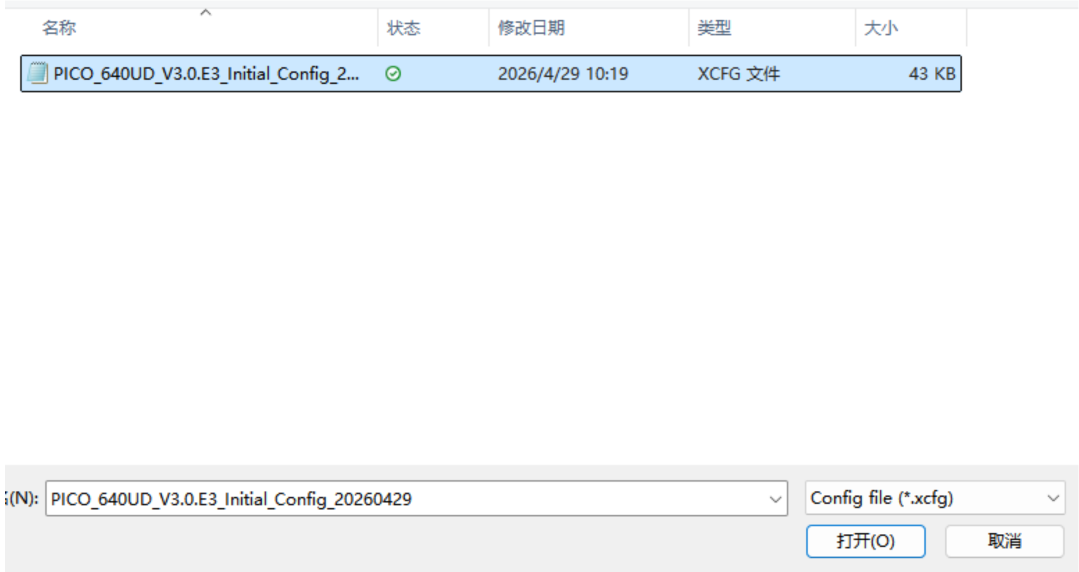
Config下载

- 点击Configuration
  - 点击Load Config



- 选择初始配置参数
  - 如附件



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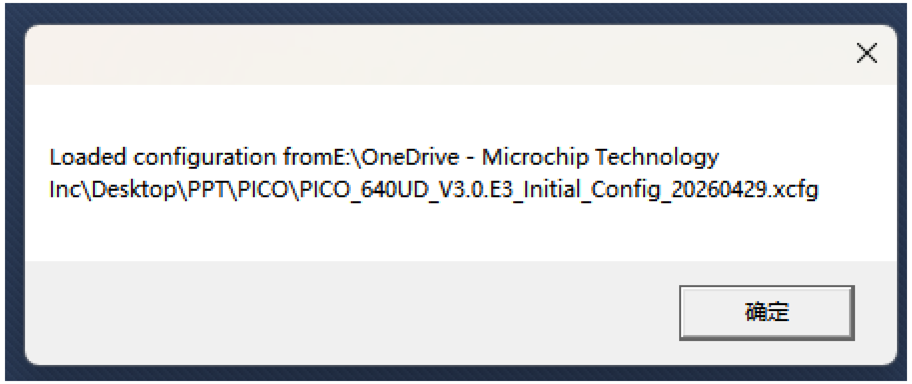
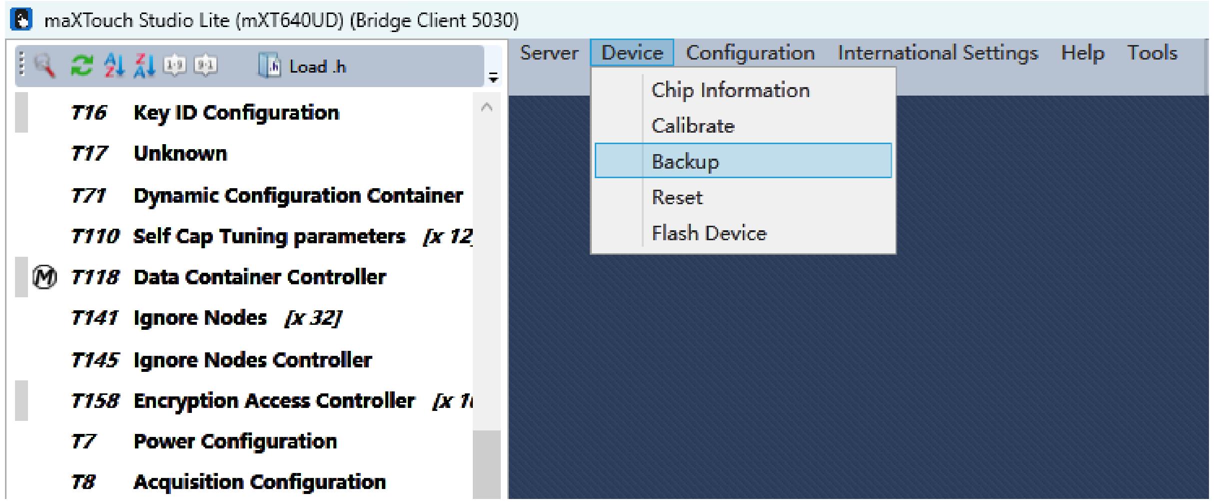
Config下载

- 下载成功会有下面提示

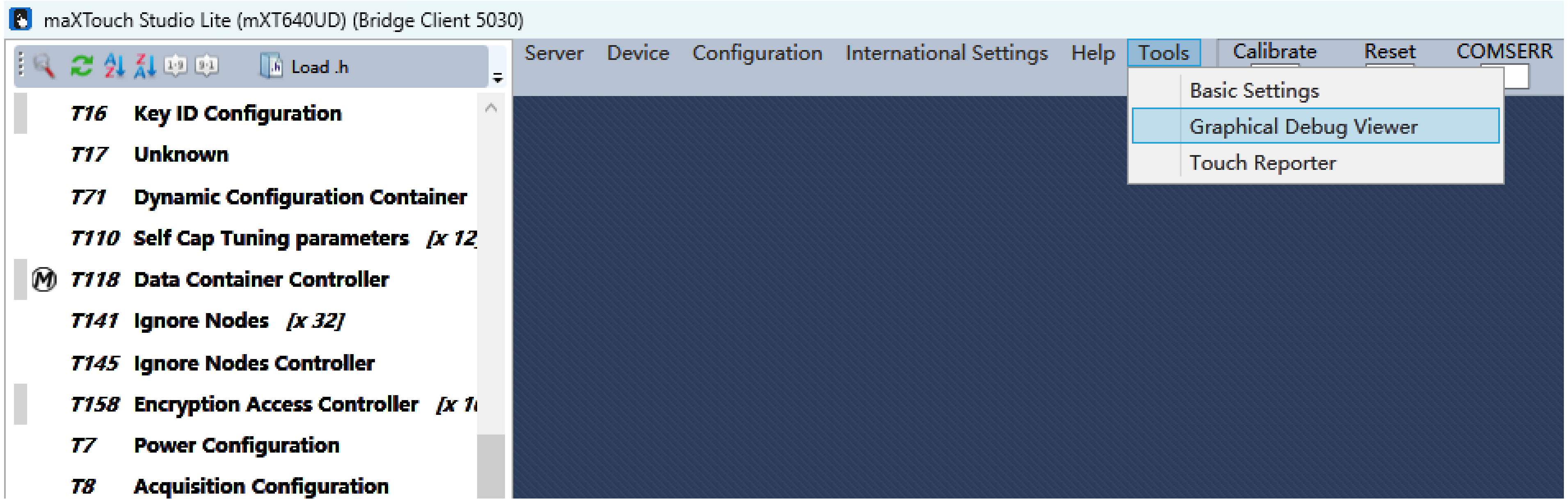


- **Backup和Reset**
  - 点击Device下的Backup
  - 再点击Device下的Reset



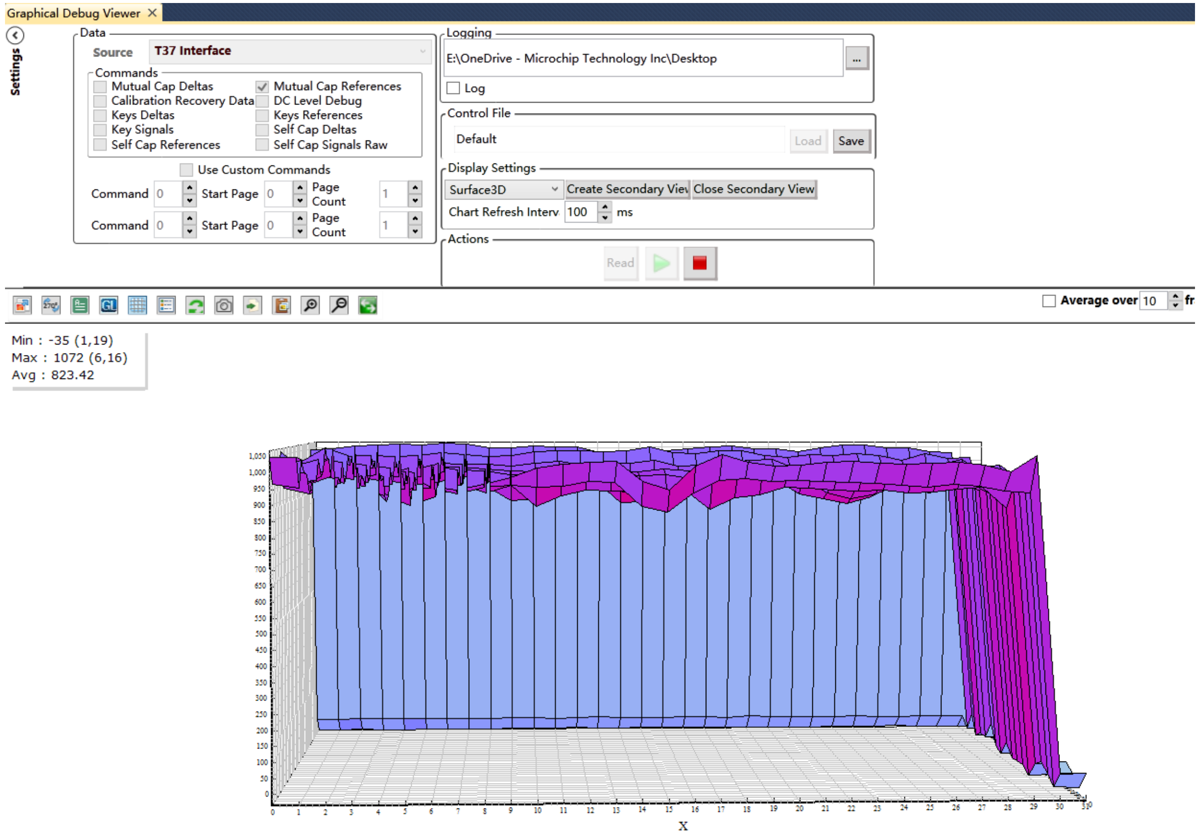
查看数据

- 打开**Tools**
  - 选择Graphical Debug Viewer

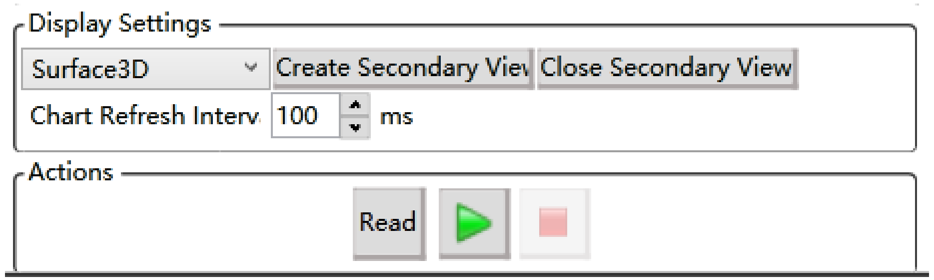


查看数据 - I2C接口

- Source选择T37 Interface



- Command选择Mutual Cap Reference
- 点击绿色的三角



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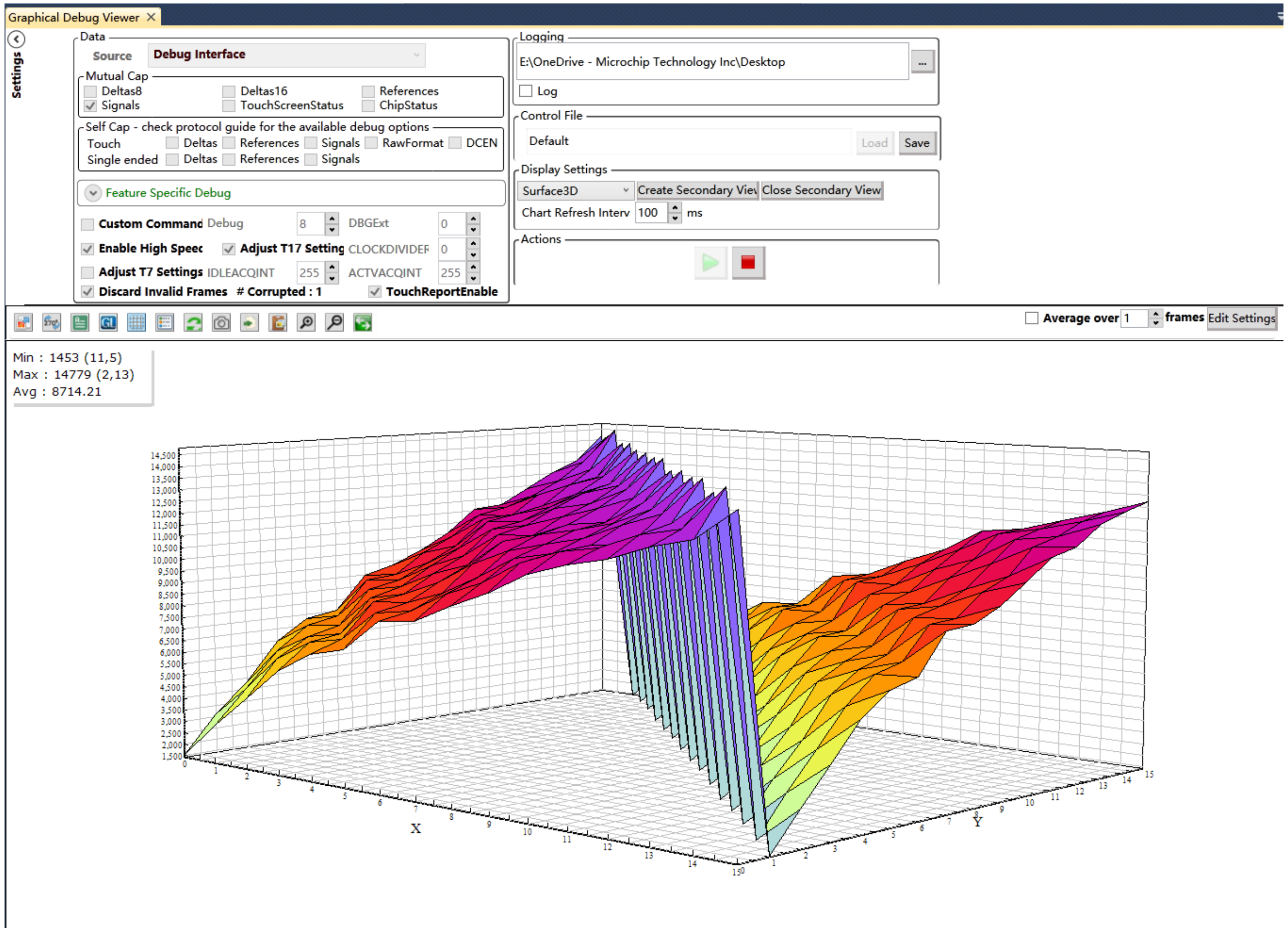
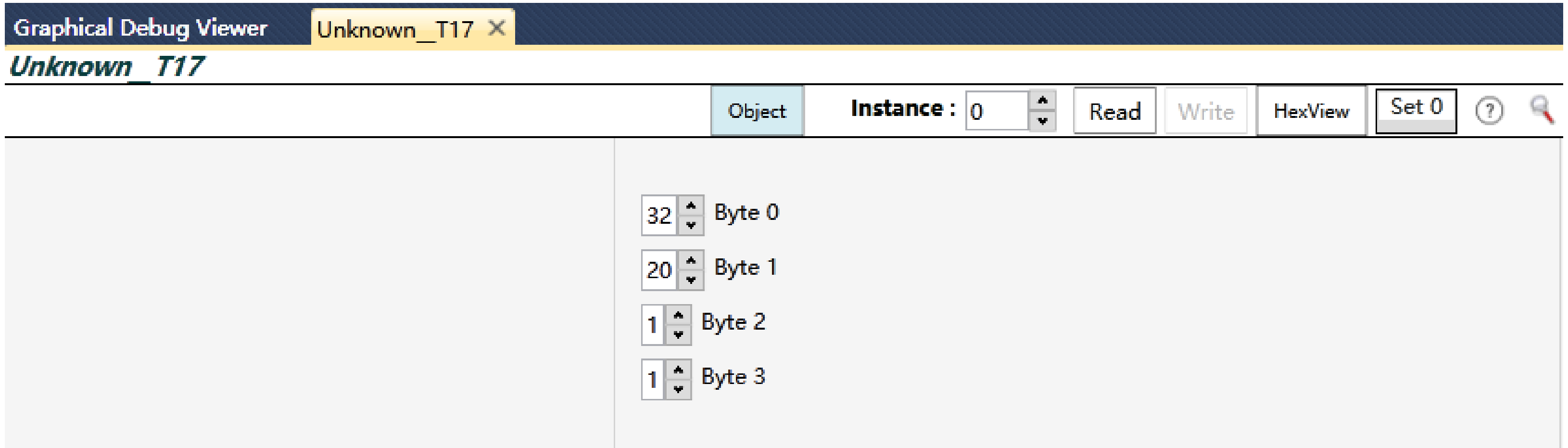
查看数据 – SPI接口

- Souce接口选择Debug Interface
- 选择Signals



怎么调整SPI速度

- **调整T17的byte1**
  - 最高60MHz



CLOCKDIVIDER Field

This field specifies the divider on the system clock used to calculate the frequency at which the SPI interface will run. The effect of this control field is shown in Figure 6-4.

Range: See Table 6-29

TABLE 6-29: CLOCKDIVIDER Field

Value	Meaning
0 (default)	Divider selected automatically to achieve the highest SPI clock frequency not exceeding 4 MHz.
1 to 255	SPI clock frequency is derived from the system clock divided by the specified value (60 MHz to 235 kHz).

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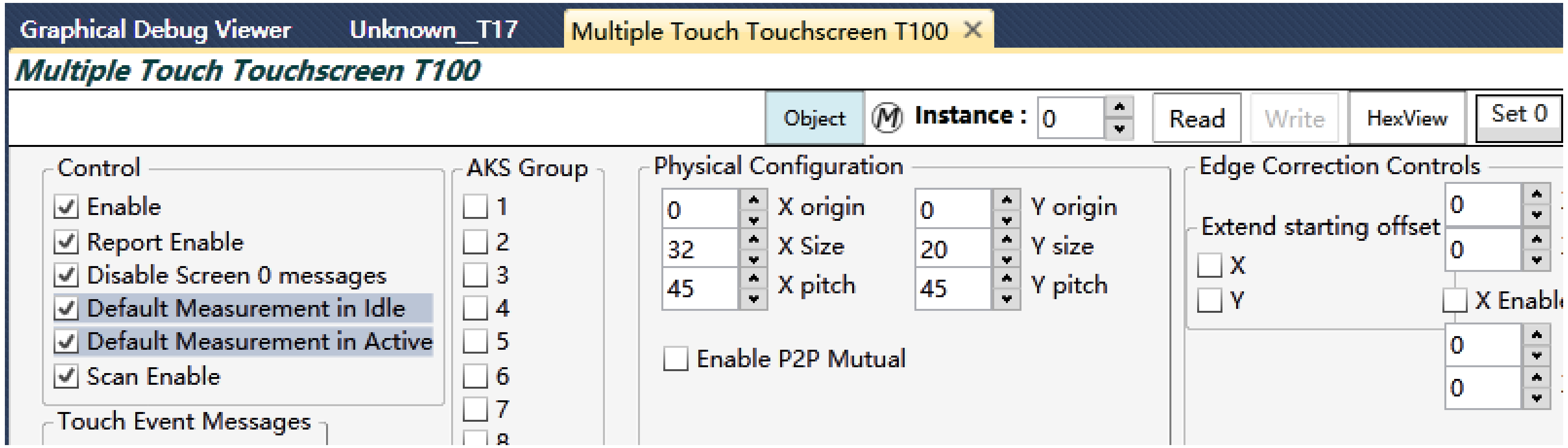
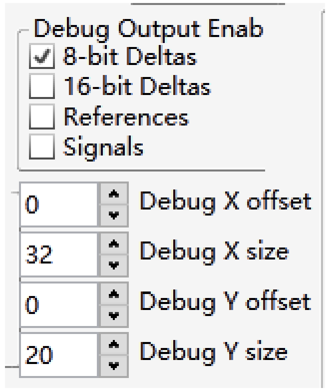
怎么调整XSize和YSize

- **T100 Physical Configuration**
  - 可以调整X Size和Y Size
  - 初始配置为32*20



怎么调整Debug Output

- **T100的参数对应如下**



怎么调整增益Gain

- **T100参数如下**
  - 只调整Gain就可以
  - 初始配置为1